TRAFFIC-LIGHT CONTROLLER

+ SYNCHRONOUS MOD-6 COUNTER

A structured design study using a Moore finite-state machine, D flip-flops, Boolean minimization, and synchronous counter logic.

TWO DESIGNS. ONE DIGITAL-LOGIC FOUNDATION.

Explore how state tables and next-state equations describe sequential digital behavior.

4	6
FSM states	count states

PROJECT OVERVIEW

01 Traffic-light controller

A four-state Moore FSM for two roads, with state assignments, output decoding, and minimized next-state logic.

MOORE FSM · D FLIP-FLOPS

02 Synchronous mod-6 counter

A three-bit counter that advances through six legal states, wraps at 101, and recovers from unused states.

NEXT-STATE LOGIC · SYNCHRONOUS

DESIGN SEQUENCES

Traffic-light FSM

S0	A: Green / B: Red
S1	A: Yellow / B: Red
S2	A: Red / B: Green
S3	A: Red / B: Yellow

Counter sequence

000 → 001 → 010 → 011 → 100 → 101

The next clock edge after 101 returns the counter to 000.

SCOPE NOTE

The traffic controller assumes two roads, no sensor inputs, and transitions triggered by an externally selected timing interval. This document presents the logic design; it does not claim a verified Verilog simulation or physical hardware implementation.

Two-road traffic-light controller

A Moore implementation is used: each lamp pattern is decoded from the registered state and changes after a clock edge.

STATE CYCLE

S0 00
Road A GREEN
Road B RED

→

S1 01
Road A YELLOW
Road B RED

→

S2 10
Road A RED
Road B GREEN

→

S3 11
Road A RED
Road B YELLOW

Cycle: S0 -> S1 -> S2 -> S3 -> S0 | Reset initializes S0 (00).

STATE / NEXT-STATE / OUTPUT TABLE

Present	Q1Q0	Next	D1D0	Road A	Road B
S0	00	S1	01	Green	Red
S1	01	S2	10	Yellow	Red
S2	10	S3	11	Red	Green
S3	11	S0	00	Red	Yellow

LOGIC EQUATIONS

NEXT-STATE LOGIC

D1 = Q1 XOR Q0

D0 = NOT Q0

For the D flip-flops, each D input is the required next-state bit.

OUTPUT DECODE

Road A
AG = NOT Q1 AND NOT Q0
AY = NOT Q1 AND Q0
AR = Q1
Road B
BG = Q1 AND NOT Q0
BY = Q1 AND Q0
BR = NOT Q1

DESIGN NOTES

Timing model: One state transition occurs per externally selected timing interval (green or yellow duration). No sensor inputs are included.

Why Moore? Lamp outputs depend only on the current registered state, helping keep each output pattern stable between clock edges.

Synchronous mod-6 counter

Three D flip-flops represent six legal states. All flip-flops share the same clock, and the next-state logic makes 101 return to 000.

COUNT SEQUENCE

000 COUNT 0 → 001 COUNT 1 → 010 COUNT 2 → 011 COUNT 3 → 100 COUNT 4 → 101 COUNT 5

Next rising clock edge after state 101: 000 (synchronous wrap-around).

COMPLETE NEXT-STATE TABLE

Present Q2Q1Q0	Next D2D1D0	Meaning
000	001	0 to 1
001	010	1 to 2
010	011	2 to 3
011	100	3 to 4
100	101	4 to 5
101	000	Terminal wrap
110	000	Unused-state recovery
111	000	Unused-state recovery

MINIMIZED NEXT-STATE EQUATIONS

D2 = SUM m(3,4)

NOT Q2 AND Q1 AND Q0
OR Q2 AND NOT Q1 AND NOT Q0

D1 = SUM m(1,2)

NOT Q2 AND (NOT Q1 AND Q0
OR Q1 AND NOT Q0)

D0 = SUM m(0,2,4)

NOT Q0 AND (NOT Q2 OR NOT Q1)

TERMINAL-COUNT CHECK

TC = Q2 AND NOT Q1 AND Q0

TC is 1 only at 101. Substituting 101 into the next-state equations gives D2D1D0 = 000, so the next rising clock edge wraps the count to 000.

RECOVERY BEHAVIOR

Unused encodings 110 and 111 both transition to 000. This makes the listed next-state logic self-recovering from those two unused states.